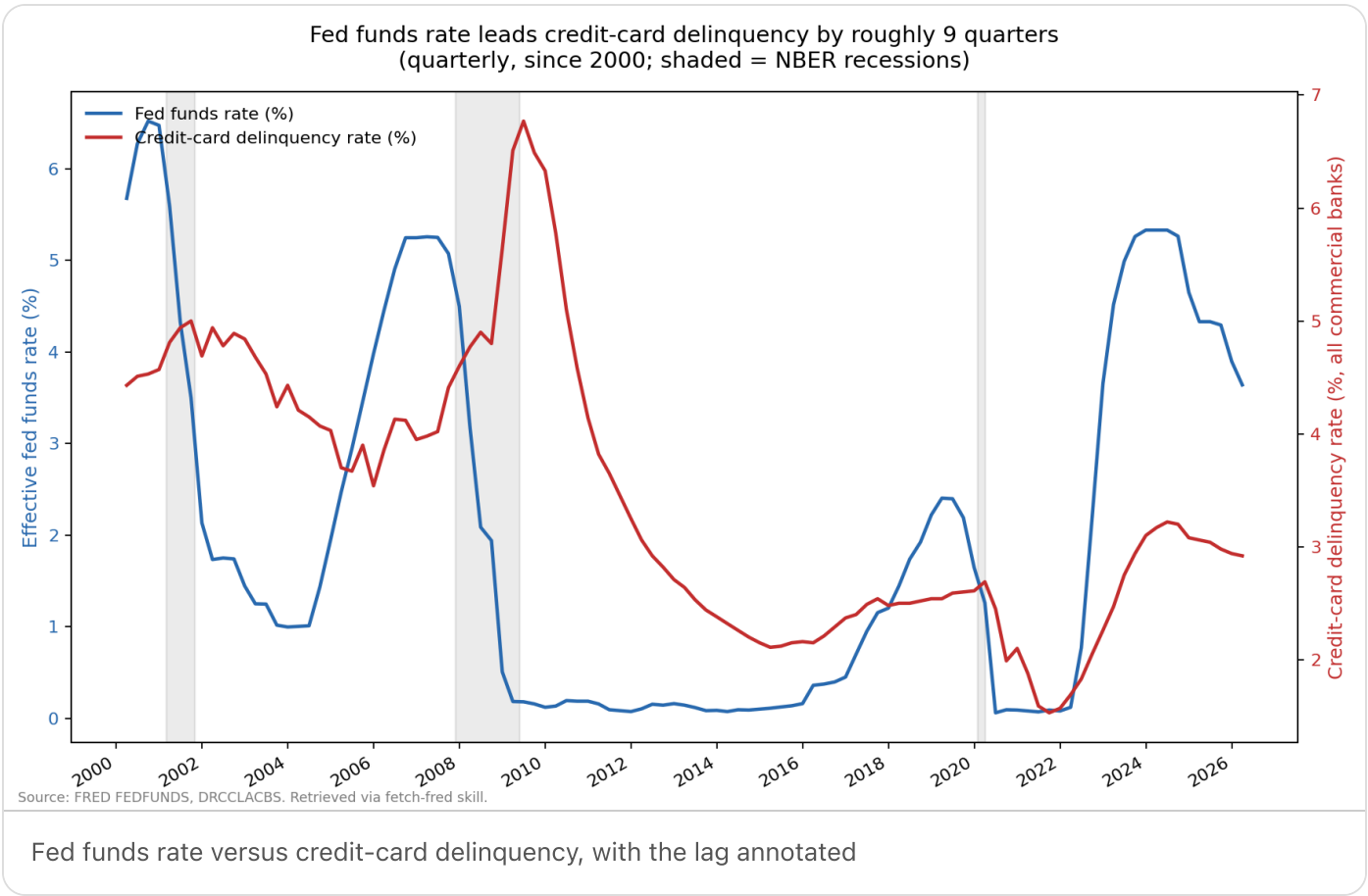


Rate Cycles and Credit-Card Delinquency: A ~2-Year Lag



The finding

Since 2000, changes in the effective fed funds rate lead changes in the credit-card delinquency rate by roughly **9 quarters (about 2 years)**. Cross-correlating quarter-over-quarter changes in each series at lags of 0–12 quarters, correlation is essentially flat (and near zero) through lag 4, then rises steadily and peaks at lag 9 ($r \approx 0.29$ on quarterly changes; $r \approx 0.69$ on levels at the same lag). This matches the visible pattern in the chart: every major hiking cycle (2004–06, 2015–18, 2022–23) is followed by a delinquency climb that shows up roughly two years later, well after the Fed has often already started cutting.

The 2008–09 spike is the clearest case: the fed funds rate topped out in mid-2007, and credit-card delinquency didn't peak until Q3 2009 — about 9 quarters later, right in line with the estimated lag.

Results

- **Data:** FRED `FEDFUNDS` (Federal Funds Effective Rate, monthly, 319 obs, 2000-01-01 to 2026-07-01, last updated by FRED 2026-08-03) and `DRCLACBS` (Delinquency Rate on Credit Card Loans, All Commercial Banks, quarterly, 105 obs, 2000-01-01 to 2026-01-01, last updated by FRED 2026-05-19). Both resampled/aligned to quarterly (fed funds averaged across each quarter's three months) via the workspace's `fetch-fred` skill.
- **Method:** quarter-over-quarter change in each series; Pearson correlation between the fed-funds change at quarter t and the delinquency change at quarter $t + lag$, for $lag = 0 \dots 12$ quarters.
- **Lag estimate:** correlation is near zero for lags 0–4 (delinquency doesn't react immediately), climbs from lag 5 onward, and peaks at **lag 9** ($r = 0.294$, $n = 95$ quarter-pairs). Level correlation at the same 9-quarter lag is much stronger ($r = 0.688$), consistent with delinquency responding to the *cumulative* level of the rate cycle rather than to any single quarter's move.
- **Coverage caveat:** `DRCLACBS` is reported roughly a quarter behind current date, per FRED's own last-updated metadata above — the most recent 1–2 quarters in the chart should be read as provisional.
- **Chart:** `fedfunds_delinquency.png` — dual-axis line chart, fed funds rate and delinquency rate since 2000, NBER recessions shaded, with the lag finding in the title.

What follows for a risk team

- **Underwriting and reserving decisions made today are reacting to a rate environment that will show up in delinquency roughly two years from now.** Tightening credit standards only after delinquency visibly rises means reacting to a rate move the Fed made ~2 years earlier — underwriting should lead the rate cycle, not the delinquency data.
- The current cycle (hikes through 2023, cuts since 2024) implies the delinquency impact of the 2022–23 hikes is still working through the book; the ~9-quarter lag puts the expected peak of this cycle's delinquency effect around late 2025–2026, which is roughly where the chart currently sits (2.9% as of the latest quarter). Reserving models should not assume delinquency has already priced in the full effect of the 2022–23 hiking cycle.
- Because the relationship is a *lag*, not a lockstep move, it is a directional planning signal — not a coefficient to hard-wire into a scoring model without further validation. Treat the ~9-quarter, ~2-year figure as an early-warning window, and revisit it as more quarters of the current cycle's data land (next `DRCLACBS` print will still trail by about a quarter, per the provenance above).

FinTechCo is fictional and this estate is a stand-in — no production system and no real customer data. The analysis uses public FRED series; the chart is the run's own output.